

## **Acme Video Rental Database Description**

The Acme Video Rental database is designed to support the operational needs of a local video store that rents out VHS tapes and DVDs. This system addresses the challenges the store previously faced with paper-based record keeping by providing a structured, relational database to track movie inventory, customer rentals, and distributor relationships. The core of the database revolves around movies and their physical copies in either DVD or VHS format. Each movie can have multiple copies, and each copy is uniquely identifiable in the inventory. To accommodate the scenario where the same movie copy can be obtained from different distributors, a bridge entity called Catalog is used to record wholesale prices and distributor-copy relationships. The database also captures the relationship between movies and film staff, including actors, actresses, and directors. A bridge entity called MovieStaff allows many staff members to be associated with each movie and defines their role. In addition, awards such as Academy Awards are tracked and can be attributed to either movies or staff members, based on the award's context. Customers and rentals are represented in such a way that every transaction is logged with associated fees (late, damage, failure to rewind), prices, and dates. This allows the store to maintain full visibility over rental activity and customer behavior. Together, these entities support detailed reporting and allow for robust querying by criteria such as actor, genre, award status, or rental history.

### **Sentence Pairs:**

#### **1. Distributor ↔ Catalog**

1. One distributor can have one or more catalog entries.
2. Each catalog entry must be associated with exactly one distributor.

## **2. Copy ↔ Catalog**

1. One copy can appear in one or more catalog entries.
2. Each catalog entry must reference exactly one copy.

## **3. Movie ↔ Copy**

1. One movie can have one or more copies.
2. Each copy must be associated with exactly one movie.

## **4. Movie ↔ MovieStaff ↔ Staff**

1. One movie can be associated with one or more staff members through the MovieStaff relationship.
2. One staff member can work on zero or many movies through the MovieStaff relationship.

## **5. Movie ↔ Award**

1. One movie can receive zero or many awards.
2. Each award may reference zero or one movie.

## **6. Staff ↔ Award**

1. One staff member can receive zero or many awards.
2. Each award may reference zero or one staff member.

## **7. Customer ↔ Rental**

1. One customer can have zero or many rentals.

- Each rental must be associated with exactly one customer.

### 8. Copy ↔ Rental

- One copy can be rented zero or many times.
- Each rental must reference exactly one copy.

#### Metadata Table:

| Acme Video Rental |                        |        |               |                                               |
|-------------------|------------------------|--------|---------------|-----------------------------------------------|
| Entity Name       | Attributes             | PK/FK? | Data Types    | Description                                   |
| Distributor       | DistributorID          | PK     | NUMBER        | Unique ID for each distributor                |
|                   | Name                   |        | VARCHAR2(100) | Name of the distributor                       |
|                   |                        |        |               |                                               |
| Catalog           | DistributorID + CopyID | PK     | NUMBER        | Unique ID for catalog entry (bridge)          |
|                   | DistributorID          | FK     | NUMBER        | References the distributor supplying the copy |
|                   | CopyID                 | FK     | NUMBER        | References the specific movie copy            |
|                   | Wholesale price        |        | NUMBER        | Price the store pays the distributor          |
|                   |                        |        |               |                                               |
| Movie             | Movie ID               | PK     | NUMBER        | Unique ID for each movie                      |
|                   | Title                  |        | VARCHAR2(150) | Title of the movie                            |
|                   | Length                 |        | NUMBER        | Movie duration in minutes                     |

|            |               |    |                  |                                                                    |
|------------|---------------|----|------------------|--------------------------------------------------------------------|
|            | Type          |    | VARCHAR2(50)     | Genre/type of the movie<br>(Horror)                                |
|            | Rating        |    | VARCHAR2(5)      | rating (PG-13, R)                                                  |
|            | Release Date  |    | DATE             | Date the movie was released                                        |
|            |               |    |                  |                                                                    |
| MovieStaff | MovieStaff ID | PK | NUMBER           | Unique ID for each movie-staff<br>relationship                     |
|            | Staff ID      | FK | NUMBER           | References the staff member<br>involved                            |
|            | Movie ID      | FK | NUMBER           | References the movie worked<br>on                                  |
|            | Role          |    | VARCHAR2(50)     | Role in the movie (Actor,<br>Director)                             |
|            |               |    |                  |                                                                    |
| Staff      | Staff ID      | PK | NUMBER           | Unique ID for each staff<br>member                                 |
|            | Name          |    | VARCHAR2(100)    | Name of the<br>director/actor/actress                              |
|            |               |    |                  |                                                                    |
| Award      | Award ID      | PK | NUMBER           | Unique ID for each award                                           |
|            | Name          |    | VARCHAR2(100)    | Name of the award (Oscar)                                          |
|            | Year          |    | NUMBER(4)        | Year the award was given                                           |
|            | Movie ID      | FK | NUMBER /<br>NULL | References the movie that<br>received the award (if<br>applicable) |

|          |              |    |                  |                                                                           |
|----------|--------------|----|------------------|---------------------------------------------------------------------------|
|          | Staff ID     | FK | NUMBER /<br>NULL | References the staff member<br>that received the award (if<br>applicable) |
|          |              |    |                  |                                                                           |
| Copy     | Copy ID      | PK | NUMBER           | Unique ID for each copy<br>(DVD/VHS)                                      |
|          | Movie ID     | FK | NUMBER           | References the movie the copy<br>belongs to                               |
|          | Price        |    | DECIMAL          | Rental price for this copy                                                |
|          | Format       |    | VARCHAR2(3)      | Format of the copy (DVD or<br>VHS)                                        |
|          | Rented       |    | BOOLEAN          | True if the copy is rented out                                            |
|          | SN           |    | NUMBER           | Serial number for inventory<br>tracking                                   |
|          |              |    |                  |                                                                           |
| Customer | Customer ID  | PK | NUMBER           | Unique ID for each customer                                               |
|          | Name         |    | VARCHAR2(100)    | Full name of the customer                                                 |
|          | Address      |    | VARCHAR2(200)    | Street address of the customer                                            |
|          | Phone Number |    | VARCHAR2(15)     | Contact phone number                                                      |
|          |              |    |                  |                                                                           |
| Rental   | Rental ID    | PK | NUMBER           | Unique ID for each rental<br>transaction                                  |
|          | Copy ID      | FK | NUMBER           | References the rented copy                                                |
|          | Customer ID  | FK | NUMBER           | References the renting<br>customer                                        |

|  |                       |  |         |                                            |
|--|-----------------------|--|---------|--------------------------------------------|
|  | Checkout Date         |  | DATE    | Date the copy was checked out              |
|  | Return Date           |  | DATE    | Date the copy was returned                 |
|  | Sale Price            |  | DECIMAL | Total cost charged for the rental          |
|  | Late Fee              |  | DECIMAL | Fee applied for late returns               |
|  | Discount              |  | NUMBER  | Discount percentage applied to this rental |
|  | Damage Fee            |  | DECIMAL | Fee for any damage to the copy             |
|  | Failure to Rewind fee |  | DECIMAL | Fee for failing to rewind VHS copy         |

